A Details of SpeechR Dataset

A.1 Reasoning Type Design

We categorize reasoning tasks in SpeechR based on three core dimensions that reflect the cognitive characteristics of reasoning: (S1) knowledge dependence, (S2) reasoning transparency, and (S3) evaluation determination.

- S1 identifies the primary knowledge source required to solve the reasoning task: factual knowledge (e.g., commonsense or background knowledge), procedural rules (e.g., mathematical or scientific laws), or normative principles (e.g., moral or behavioral norms).
- S2 evaluates the level of reasoning transparency, i.e., whether solving a task requires intermediate steps such as logical deduction or chain-of-thought reasoning.
- S3 evaluates whether a task has a clearly objective ground-truth answer, as opposed to relying on subjective judgment.

Based on these dimensions, we define three reasoning types in SpeechR:

- Factual Reasoning involves understanding and retrieving factual or commonsense information. These tasks do not require multi-step reasoning and are evaluated using objective criteria. Examples include "Why are cats afraid of water?" or "Where did you go yesterday?", which rely on language comprehension and real-world knowledge.
- Procedural Reasoning requires explicit, multi-step logical or numerical inference. Tasks in this category often involve deterministic reasoning chains and verifiable outputs, such as "What is 12 divided by 3 plus 5?" or "If A is taller than B, and B is taller than C, who is the tallest?".
- Normative Reasoning involves assessing actions, intentions, or social behaviors based on implicit moral or social norms. These tasks may involve open-ended answers and often require subjective evaluation, reflecting the inherent ambiguity in social and ethical reasoning. Examples include "Is this message likely to be a scam?" or "Was second speaker's behavior ethically acceptable?".

In summary, SpeechR adopts a three-type taxonomy consisting of factual, procedural, and normative reasoning, based on cognitive criteria (S1 to S3), enabling comprehensive evaluation of LALMs' reasoning breadth and depth in speech-based scenarios.

A.2 Readability Enhancement

We apply a set of controlled rewriting rules to ensure that the speech is syntactically well-formed, semantically faithful, and suitable for high-quality audio synthesis. Specifically, we adopt the following criteria:

- The text length is constrained to fall within 7 to 150 words.
- All emojis and special characters (e.g., ":)" or "#") are removed.
- Abbreviations and incomplete words are expanded into their full forms (e.g., "tkts" becomes "tickets", "comp" becomes "competition").
- URL links are simplified to include only the domain name.
- Appropriate punctuation and sentence segmentation are added to improve clarity and naturalness when read aloud.
- No new content is introduced beyond what is present in the original input.

A.3 Interaction Enrichment

To simulate natural conversational dynamics, we enhance the interactivity of the original text-based datasets using two strategies: restructuring the data format and modifying pronoun perspectives.

Restructuring the Data Format. Some source datasets do not follow a dialogue or question-answer format. For these, we convert each instance into a unified conversational structure.

For example, in the DailyDilemmas dataset, each instance includes a moral scenario and the outcomes of different actions (e.g., choosing to act or not). We construct the **Person A** utterance by combining the scenario and a corresponding action-related question, and generate the **Person B** response by combining the selected action and its consequence. This transformation results in a coherent two-turn dialogue.

Modifying Pronoun Perspectives. To make the interaction feel more natural, we insert personal pronouns into the conversation. Specifically:

- The **Person A** prompt is rephrased to include "you" (e.g., "What should one do in this situation?" → "What should you do in this situation?").
- The **Person B** response is rephrased to include "I" (e.g., "One should avoid this." → "I would avoid this.").

These adjustments ensure that the resulting dialogue better mirrors speaker-centric, conversational speech patterns.

A.4 Annotation of Acoustic Features

To enrich SpeechR with acoustic cues relevant to reasoning, we annotate each instance with two types of features: stress and emotion. Annotations are generated using GPT-4o under structured prompt templates, as described below.

Stress Annotation. We provide GPT-4o with the transcribed text and the ground-truth answer, asking it to identify the keyword or phrase within the transcription that most strongly determines the correct answer. The model is explicitly instructed to select the word or phrase that should be emphasized if the sentence were spoken aloud.

Emotion Annotation. For emotion annotation, GPT-4o is given the transcribed text, the ground-truth answer, and a predefined list of emotions. It is instructed to select the emotion most appropriate for reading the text aloud, considering both the tone of the content and the underlying reasoning intent.

These annotations are used to assess whether LALMs can benefit from or are sensitive to prosodic and emotional features during reasoning tasks.

A.5 Volunteers Details

The recordings are produced by 10 volunteers (female:male = 6:4), aged between 10 and 30, with diverse nationalities including Australia, the United States, Japan, Korea, China, France, Russia, and Morocco. Volunteers are instructed to read in relatively quiet environments, while natural phenomena such as hesitations, pauses, misreadings, or slight omissions are allowed.

A.6 Comparison with Existing Benchmarks

We identify four major limitations in existing audio datasets:

(1) Most are designed for low-level audio understanding tasks such as event classification or speech recognition, rather than high-level reasoning. For example, MELD and IEMOCAP primarily target emotion recognition, while VoxCeleb focus on speaker identification.

(2) Many datasets fail to simulate realistic dialogue reasoning scenarios, limiting their applicability to natural conversational inference. For instance, MMAU provides an audio dialogue and asks the LALM to identify the roles of the speakers. However, such reasoning is relatively straightforward and lacks inferential depth. In most human or human-machine conversations, speaker roles are either explicitly stated or easily inferred from surface cues. Therefore, it fails to capture the complexity required for evaluating real-world reasoning.

(3) Existing datasets often lack diversity in reasoning types. For example, temporal reasoning and content-based reasoning benchmarks focus narrowly on specific inference categories, limiting their ability to provide a comprehensive assessment of the reasoning capabilities of LALMs.

(4) Even among reasoning-oriented datasets, the complexity of reasoning remains constrained. For instance, although MMAU introduces dialogue-like audio data, its short average audio length (approximately 10 seconds) significantly limits the depth of reasoning it can support.

In contrast, our SpeechR benchmark addresses these limitations by offering a broader range of reasoning tasks, spanning factual, procedural, and normative dimensions. It incorporates more realistic and context-rich dialogue scenarios, simulating natural conversational settings. Furthermore, SpeechR supports diverse output formats, including both multiple-choice and generative responses, and emphasizes clearer, more complex reasoning chains that reflect the multi-step inference required in real-world audio interactions. These features enable a more comprehensive and challenging evaluation of LALMs' reasoning capabilities.

A.7 Human Verification for Data Quality

To verify the reliability of the benchmark annotations, we conducted a manual inspection on a representative subset of the test set.

We randomly selected 100 examples from the main generative test split, evenly covering factual, procedural, and normative tasks. Each example included the input speech, its transcription, the gold label, and the model-generated answer. Three annotators with NLP backgrounds independently reviewed all samples and checked:

- whether the gold label correctly matched the spoken content;
- whether the transcription accurately reflected the audio.

On average, the benchmark results as Table 6 indicates that the dataset labels and transcriptions are highly reliable.

Metric	Value
Label correctness	93%
Speech–text consistency	96%

Table 6: Manual verification results for a subset of the generative benchmark.

A.8 Audio Human-Likeness

Although the audio in SpeechR is synthesized, the use of the Azure Speech SDK ensures naturalness and high-quality in the generated speech. We assessed the human-likeness of the synthesized audio by using perceptual evaluation with native listeners. We conducted a listening test with 10 native English speakers. Each participant was asked to rate 30 randomly sampled audio on a 5-point human-likeness scale (1 = robotic and unnatural, 5 = indistinguishable from natural human speech). The resulting average score was 4.8, indicating that the generated audio in SpeechR is highly natural and suitable for evaluating spoken language understanding in LALMs.

A.9 Human Evaluation Consistency

Three annotators independently rated 100 randomly sampled generative outputs from GPT-4o-audio model, balanced across procedural (CoT-based) and normative (moral or social) reasoning tasks. Pairwise consistency was computed using the Pearson Correlation Coefficient (PCC) for each generative evaluation metrics, and we report the average correlation across annotator pairs as Table 7.

Task Type	FC	LR	Coh
Procedural reasoning	0.83	0.79	0.75
Normative reasoning	0.81	0.76	–

Table 7: Average pairwise Pearson correlation among annotators for human evaluation of generative outputs.

All correlation values exceed 0.75, indicating strong agreement among annotators. These results confirm that the generative evaluation scores are stable for both procedural and normative reasoning tasks.

A.10 Subset Filtering and Analysis of Math Tasks

During preliminary evaluation, we observed unusually high accuracy in the mathematical reasoning subset of these models, especially for the generative test set (mainly derived from GSM8K-style problems). In contrast to other categories, where performance ranged between 30–60% on multiple-choice, models achieved 85–100% correctness on the math subset when chain-of-thought reasoning was enabled.

Such a large margin suggests that these items may not reflect the general reasoning difficulty of SpeechR. Given that GSM8K is widely used in model pretraining and instruction tuning, we cannot exclude the possibility that certain problems or templates overlap with materials seen during training. To avoid confounding effects from potential memorization, we exclude this subset when reporting the main results marked with an asterisk (*).

This exclusion does not affect the relative ranking of models on other reasoning categories but ensures a fairer comparison across unseen reasoning types.

B Model Descriptions

LTU. LTU is a multimodal large language model designed for general audio understanding. Trained on the OpenAQA-5M dataset, it demonstrates strong performance in audio classification and captioning tasks.

GAMA. GAMA is a general-purpose large audio-language model that integrates multiple types of audio representations. Fine-tuned with the CompA-R dataset, it enhances complex audio reasoning abilities, outperforming other LALMs in diverse audio understanding tasks.

GAMA-IT. An instruction-tuned variant of GAMA, GAMA-IT is designed to improve performance in open-ended audio question-answering tasks requiring complex reasoning. It leverages instruction tuning to enhance its reasoning capabilities.

Mellow. Mellow is a compact 167M parameter audio-language model optimized for reasoning tasks. It takes in two audio inputs and a text prompt, producing free-form text outputs. Despite its small size, Mellow achieves competitive performance with significantly fewer resources.

SALMONN. SALMONN is a large language model enabling speech, audio events, and music inputs. It supports various tasks, including automatic speech recognition, emotion recognition, and audio question-answering.

Qwen-Audio-Chat. Qwen-Audio-Chat is a multimodal model that accepts diverse audio inputs and text. It is designed for tasks such as speech recognition and audio-text understanding, emphasizing instruction-following capabilities.

Qwen2-audio-7B. An updated large-scale audio-language model, Qwen2-Audio-7B is capable of handling various audio signals and performing audio analysis or direct textual responses.

Qwen2-audio-Instruct. This is an instruction-tuned version of Qwen2-Audio-7B, enhancing the model's ability to follow prompts and perform complex reasoning tasks.

LLama-Omni. LLaMA-Omni is a speech-language model supporting low-latency, high-quality speech interactions. It can generate both text and speech responses directly from speech instructions with extremely low latency.

Qwen2.5-Omni. Qwen2.5-Omni is a fully integrated multimodal model that natively processes text, vision, audio, and video streams. It supports real-time dialogue, cross-modal translation, and synchronized perception-generation tasks, making it suitable for interactive AI agents and multimodal reasoning applications.

GPT-4o-audio-preview. OpenAI's GPT-4o-Audio-Preview is a multimodal model integrating real-time audio, vision, and text processing capabilities. It enables natural speech interactions and multilingual translation, allowing users to talk to ChatGPT with real-time responses and interruptions.

Gemini-1.5-pro. Gemini-1.5-pro is an advanced multimodal model supporting text, code, image, audio, and video inputs. It is designed for high-efficiency reasoning and generation tasks, with capabilities in understanding and interacting with various data modalities.

Gemini-2.5-Pro. Gemini-2.5-Pro is a large-scale foundation model with extended context and advanced multimodal reasoning. It unifies text, code, image, audio, and video understanding, enabling complex analytical workflows and high-fidelity multimodal generation across diverse domains.

C Source Datasets

In this section, we introduce the set of datasets utilized in the construction of the SpeechR benchmark.

ReClor. ReClor is a reading comprehension dataset consisting of logical reasoning questions derived from standardized graduate-level entrance exams. Each instance includes a passage, a question, and multiple-choice answers, with a strong focus on deductive reasoning.

BoolQ. BoolQ is a yes/no question-answering dataset where each question is paired with a short supporting passage. The questions are naturally occurring and require understanding of factual content from the given context.

CommonsenseQA. CommonsenseQA is a multiple-choice question answering dataset that targets commonsense reasoning. It is built on ConceptNet relations and presents challenging examples that often require reasoning beyond surface-level word matching.

RiddleSense. RiddleSense is a dataset designed to evaluate lateral thinking and creative commonsense reasoning. It contains multiple-choice riddles, where the correct answer requires both semantic understanding and reasoning through implicit associations.

GSM8K. GSM8K is a math word problem dataset designed to assess grade-school level arithmetic reasoning. It includes detailed step-by-step chain-of-thought annotations, making it a standard benchmark for evaluating procedural reasoning in CoT settings.

ReveAL-CoT. ReveAL-CoT is a scientific reasoning dataset featuring multi-step inference questions across physics, biology, and other science domains. Each question is annotated with chain-of-thought explanations to support structured procedural reasoning.

ETHICS. ETHICS is a benchmark for moral reasoning that includes scenarios requiring judgments about the ethical permissibility of actions. The dataset covers diverse moral contexts such as fairness, harm, and loyalty.

DailyDilemmas. DailyDilemmas contains short narratives that describe everyday situations involving social or ethical decision-making. Each instance asks the model to evaluate the appropriateness or morality of an individual's behavior.

SMS Spam Collection. This dataset contains a collection of labeled SMS messages, with each message categorized as spam or ham (not spam). It is widely used for binary classification tasks involving deception or malicious intent detection.

Enron Email. The Enron Email dataset comprises real-world corporate email communications, with a subset labeled for spam detection. It supports studies in behavioral and normative analysis, especially in identifying unethical or misleading content.

D Prompt Templates

In this section, we present the prompt templates used throughout various stages of our dataset construction and evaluation pipeline. Figures 4, 5, and 6 illustrate three prompt designs employed during the generation of the SpeechR dataset. Specifically, Figure 4 shows the prompt used to enhance readability and linguistic fluency of raw samples, Figure 5 demonstrates the interaction-oriented prompt that encourages more engaging and context-aware formulations, and Figure 6 presents the filtering prompt used for quality control, enabling the exclusion of incoherent or irrelevant data.

In addition, Figures 7 and 8 display prompt templates used in the evaluation phase. Figure 7 is designed for emotion annotation and highlight word extraction from audio transcripts, and is applied specifically to the mini version of SpeechR. Figure 8 illustrates the prompt format adopted for LLM-as-a-judge evaluation of the generative version, guiding the model to assess response correctness, logical relevance, and reasoning coherence.

E Ablation Study

We conduct ablation studies to clarify two key questions frequently raised in the evaluation of speech-based reasoning benchmarks: whether LALMs genuinely perform audio-conditioned reasoning beyond transcription.

E.0.1 Text-only and ASR-to-Text Baselines. A common concern is whether the performance of large audio-language models can be reduced to text reasoning after transcription. To disentangle transcription effects from reasoning ability, we compare end-to-end LALMs with text-only and ASR-to-text baselines, as shown in Table 9.

Across all reasoning categories, both text-only and ASR-to-text baselines substantially outperform the end-to-end LALM, with particularly large gaps in factual and normative reasoning. This indicates that reasoning over spoken input remains significantly more challenging than reasoning over text, even when the underlying semantic content is identical. Moreover, the strong performance of the ASR-to-text pipeline suggests that current LALMs are not equivalent to a simple ASR+LLM cascade; instead, the primary bottleneck lies in speech-conditioned reasoning rather than transcription accuracy.

These results demonstrate that SpeechR does not merely evaluate text reasoning in disguise, but instead exposes reasoning challenges unique to spoken input, highlighting substantial headroom for future LALM development.

F Qualitative Analysis

As shown in Table 8, we present qualitative results from different LALMs across the three reasoning categories in SpeechR: factual reasoning, illustrated with creative puzzles; procedural reasoning, represented by mathematical problems; and normative reasoning, which highlights the models' ability to generate inferences in dialogue-based scenarios.

G Reflections and Future Directions

SpeechR is designed as an initial step toward evaluating reasoning in speech-based interactions. While it covers a diverse range of reasoning tasks and introduces controlled acoustic variations, further extensions may broaden its scope in the following directions:

Speech Variability. All speech in SpeechR is consistently synthesized using standardized settings. Expanding to include more varied prosody, speaking styles, and spontaneous speech could offer richer insights into real-world model performance.

Linguistic and Cultural Coverage. Current data construction focuses on English with general social contexts. Exploring additional languages and sociocultural scenarios could enable broader applicability across multilingual and multicultural settings.

Interaction Dynamics. The current benchmark emphasizes static single-turn prompts. Incorporating multi-turn dialogue and speaker dynamics could allow future benchmarks to capture more interactive aspects of speech-based reasoning.

(Factual Reasoning)
There was a green house in the green house was a white house in that white house was a red house in that red house where eggs. What is it?
A: salt B: jar C: the park D: supermarket E: watermelon
Ground Truth: E

LTU: The best option is A. salt.
GAMA: The best option is D. supermarket. The audio mentions 'the park' and 'watermelon', but not 'supermarket'.
GAMA-IT: The audio does not provide enough information to accurately determine the location.
Mellow: a particular way is best said about the best option...
SALMONN: E. watermelon
Qwen-Audio-Chat: The best option is C. the park.
Qwen-Audio-7B: what is it?
Qwen-Audio-Instruct: It is a riddle that describes a series of houses based on their colors. The answer to the riddle is Egg
LLaMA-Omni: The correct answer is B. jar
GPT-4o: The riddle describes a watermelon. The green house refers to the green outer rind, the white house refers to the white inner layer, the red house refers to the red flesh, and the eggs refer to the seeds inside. Therefore, the best option is E. watermelon.
Gemini-1.5-Pro: The riddle describes a watermelon. Therefore the answer is **E. watermelon**.

(Procedural Reasoning)
John takes care of 10 dogs. Each dog takes .5 hours a day to walk and take care of their business. How many hours a week does he spend taking care of dogs?
A: 65 B: 35 C: 32.5 D: 30
Ground Truth: B

LTU: B.35
GAMA: A.35
GAMA-IT: The correct option is D.30
Mellow: A
SALMONN: B.35
Qwen-Audio-Chat: The answer is A.65.
Qwen-Audio-7B: how many hours a week does he spend taking care of dogs?
Qwen-Audio-Instruct: John spends 7 hours a week taking care of dogs.
LLaMA-Omni: John spends fifty hours a week taking care of dogs.
GPT-4o: B.35
Gemini-1.5-Pro: B

(Normative Reasoning)
Person A: You're a project manager in a marketing firm. You notice a colleague using company resources for a personal project during office hours, undermining the company's policy. Should you take initiative and report this to your superior?
Person B: Yes. I report the colleague to my superior. The colleague is reprimanded and harbors resentment towards me, creating a hostile work environment. This affects team morale and productivity, leading to a decline in the quality of work.
Question: What moral values are reflected in this response?
Ground Truth: "respect for rules", "accountability", "responsibility", "honesty", "trust", "integrity", "fairness"

Mellow: The response from the other woman is a brief, soft utterance, likely a comment...
SALMONN: The speaker values taking initiative and reporting unethical behavior to their superior.
Qwen-Audio-Chat: The second speaker mentions that they value honesty and integrity in the workplace.
Qwen-Audio-7B: [Music]
Qwen-Audio-Instruct: I cannot provide a definitive answer as it depends on various factors such as company policies, cultural norms, and individual personalities.
LLaMA-Omni: I would take initiative and report this to my superior.
GPT-4o: The speaker values adherence to company policies and is willing to take action to uphold them, even if it may lead to personal or team conflict
Gemini-1.5-Pro: The speaker prioritizes upholding company policy and addressing misconduct, even at personal cost.

Table 8: Examples of responses from LALMs across different reasoning categories. Blue text indicates abnormal or incorrect responses.

Readability Enhancement

#Take SMS as an example.

You are a speech-content analyzer.
You are a professional editor specialized in cleaning and improving the readability of messy SMS messages.

Given a disorganized, advertisement-like SMS text, your task is to:
Correct grammar and expand incomplete words (e.g., "tkts" → "tickets", "comp" → "competition").
Add appropriate punctuation and sentence breaks to make the text easy and natural to read aloud.

Preserve the original meaning and promotional intent of the message.
Do not invent or add any new information.

Format the output into short sentences or clear bullet points if it improves flow and readability.

If any emoticons (e.g., ":)") are present, remove them directly.
Keep the overall length and structure of the original message as much as possible.
Goal:
Make the message sound natural, clear, and smooth when read aloud.

Here is the SMS to improve:
Please output only the cleaned and improved version of the message..

#In-context examples

[Output format]

Figure 4: Prompt template for dataset readability enhancement.

Interaction Enrichment

#Take Email as an example.

You are a professional email content extractor and security analyst.

You are given an email. Please extract only the email content — exclude metadata such as Subject, Sender, Receiver, Date, etc.

Then, analyze the extracted content and determine whether the email is classified as "normal", "spam or potential threat" based on its content and any security risks.

The final output format should be:
messageid: [The message ID of the email]

emailcontent: [The extracted pure content of the email]

label: [normal / spam or potential threat]

Requirements:
Focus only on the main body text of the email.

Make a judgment based solely on the extracted content.

Ensure the output strictly follows the required format.
#In-context examples

[Output format]

Figure 5: Prompt template for dataset interaction enrichment.

Dataset filtering

Task: Determine if the following dialogue meets the criteria and return a JSON result with keys `valid` and `index`.

Dialogue:
{question}
Index:
{index}
Selection Criteria:

1. Exclusion of Special Symbols:
- The dialogue must not contain mathematical formulas (e.g., $\frac{{1}}{{2}}$), special symbols (e.g., ≠, ≈, ∑, →, :), or structured formats like tables or bullet points (e.g., "1. ... 2. ...").
- If the dialogue includes visual references (e.g., "Based on the image below"), it should be rejected.
2. Text Length Control:
- The dialogue text should be within **70 words**.
- Estimated audio duration should be ≤30 seconds (calculated at 150 words per minute).
3. Language Simplicity:
- Prefer dialogue set in everyday scenarios (e.g., shopping, time calculations), avoiding complex terms (e.g., "quadratic equation").
- Sentences should be clear, without nested clauses.
4. Logical Consistency:
- The dialogue content should clearly adhere to or clearly violate moral principles. The moral principles include honesty, respect, justice, and responsibility.
- Dialogues that do not clearly adhere to or violate moral principles should be rejected.
5. **Audio Feasibility**:
- The dialogue content should be suitable for audio presentation without the need for visual elements or diagrams.
- It should be easy to understand when listened to as audio, without reliance on punctuation or special formatting for clarity.

#In-context examples

[Output format]

Emotion Annotation and Highlight Extraction from Audio Transcripts

You are a speech-content analyzer.
Given a snippet of dialogue or context, you must:

1. Identify exactly 1–3 **words**, **short phrases**, or **brief clauses** (up to ~20 words each) to emphasize when spoken, and return them as an array in the "highlight" field.

2. Assign exactly one emotion label from:
['angry', 'cheerful', 'excited', 'friendly', 'hopeful', 'sad', 'shouting', 'terrified', 'unfriendly', 'whispering']

3. If only one speaker is present, judge that speaker's emotion; if two speakers, judge **only Person B**'s emotion.

Important constraints:
- **Output strictly one** JSON object with **only** the keys `"highlight"` and `"emotion"`.
- The `"highlight"` array may contain **single words**, **phrases**, or **short clauses** (not full paragraphs).
- **Do not** emit any extra text, explanation, or markdown—**only** the JSON.

#In-context examples

[Output format]
```json
{
  "highlight": ["raining", "outside"],
  "emotion": "hopeful"
}
```


Figure 7: Prompt template for emotion annotation and highlight extraction from audio transcripts for SpeechR mini version.

LLM verifier for SpeechR generative version evaluation

You are an expert evaluator. Given the question, model prediction, and its chain-of-thought (if any), provide the following metrics in JSON:
1. final_correct: For yes/no questions (reference is "Yes" or "No"), interpret the overall stance in the prediction and compare. For classification/list tasks, 1 if the prediction mentions at least one reference item or is semantically equivalent; else 0.
2. logic_relevance: an integer from 1 to 5 indicating how strongly the prediction logically follows the question:
    1 = no relevance
    2 = very weak relevance
    3 = moderate relevance
    4 = strong relevance
    5 = very strong relevance
3. cot_coherence_score: an integer from 1 to 5 assessing the chain-of-thought coherence (only if provided):
    1 = no coherence or no chain-of-thought(disjointed, illogical)
    2 = low coherence (many gaps)
    3 = moderate coherence (some gaps)
    4 = high coherence (clear, minor issues)
    5 = excellent coherence (very clear, logical flow)

Respond exactly with:
{{"final_correct":<0 or 1>,"logic_relevance":<1-5>,"cot_coherence_score":<1-5>}}

Question: {question}
Reference Answer: {answer}
Model Prediction: {prediction}

#In-context examples

Illustrated outputs are shown below:
[Output format]

Figure 8: Prompt template for LLM-Based evaluation of the SpeechR generative version.

Model	Factual Reasoning			Procedural Reasoning		Normative Reasoning				Avg
	RC	CS	CreaT	M–CoT	G–CoT	MJ	BA	SMS	Email	
Text-only	82.86	79.14	63.07	26.91	58.50	57.50	33.14	54.61	75.48	57.43
ASR-to-Text	83.12	78.78	63.34	27.54	63.00	53.13	36.98	52.67	71.94	56.86
End-to-End LALM	50.13	36.69	16.44	25.21	39.00	47.00	12.72	39.56	32.26	33.90

Table 9: Accuracy (%) comparison between text-only, ASR-to-text, and end-to-end large audio-language model (LALM) baselines. The text-only baseline uses Qwen2-7B-Instruct, the ASR-to-text baseline uses Whisper-Large-V3 to Qwen2-7B-Instruct, and the end-to-end LALM uses Qwen2-Audio-Instruct.